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Week 2: Camera Calibration

Ungraded Practice: 2.1

Overview of Camera Calibration

4 min

Practice Assignment: 2.1

Overview of Camera Calibration Self-Check Quiz

Submitted

Reading: 2.2 Video

Overview

10 min

Ungraded Practice: 2.2

Linear Camera Model

20 min

Practice Assignment: 2.2

Linear Camera Model Self-Check Quiz

Submitted

Ungraded Practice: 2.3

Camera Calibration

8 min

Practice Assignment: 2.3

Camera Calibration Self-Check Quiz

2h

Ungraded Practice: 2.4

Intrinsic and Extrinsic Matrices

8 min

Practice Assignment: 2.4

Intrinsic and Extrinsic Matrices Self-Check Quiz

2h

Ungraded Practice: 2.5

Image Points

7 min

Practice Assignment: 2.5

Image Points Self-Check Quiz

2h

Discussion Prompt: Week 2

Camera Calibration

10 min

Discussion Prompt: Week 2

Questions and Feedback

2 Questions and Feedback

Week 2 Quiz

Your grade: 100%

Your best: 100% • Your highest score

2.2 Linear Camera Model Self-Check Quiz

Next item →

1. Please select all the correct statements from the following:

1 / 1 point

☒ Obtaining camera coordinates from the world coordinates refers to coordinate transformation.

COOCH

Help me practice

Let's chat

Coordinate transformation, in the context of computer graphics and computer vision, refers to the process of converting points or coordinates from one coordinate system to another.

☐ Obtaining camera coordinates from the world coordinates refers to perspective projection.

☐ Obtaining image coordinates from the camera coordinates refers to coordinate transformation.

How is it done?

perspective

Try again

☒ Obtaining image coordinates from the camera coordinates refers to perspective projection.

☒ Correct

Your grade

To get you used at least 80%, we keep your highest score.

100%

View submission

See feedback

2. Please select all the correct statements from the following:

1 / 1 point

☐ Equations for perspective projection are cubic.

☒ Linear

☐ Bilinear

☐ Report as issue

The value of W in the homogeneous representation of coordinate points can never be zero.

☒ Correct

With homogeneous representation of coordinates is a fictitious value that is used for scaling or normalization purposes and hence, can never be zero.

☐ $[K]$ is the correct empirical representation of intrinsic matrix where K is a fundamentally correct calibration matrix.

☒ Calibration Matrix is always an upper-triangular matrix.

☒ Correct

The empirical representation of a calibration matrix, K is upper-triangular in nature:

$$K = \begin{bmatrix} \alpha_0 & 0 & \alpha_2 \\ 0 & \alpha_1 & \alpha_3 \\ 0 & 0 & 1 \end{bmatrix}$$

☒ If R is the rotation/orientation matrix of the camera in the world coordinate frame, then $R^T R = I$, where I is an identity matrix of the appropriate dimensions.

☒ Correct

The rotation/orientation matrix, R , is an orthonormal matrix. Therefore, the property $R^T R = I$ holds true.

https://www.coursera.org/learn/3d-reconstruction-multiple-viewpoints/assignment/submission/WDAW62-24linear-camera-model-self-check-quiz/view-feedback

1/1